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MACROMOLECULAR COMPOUNDS AND POLYMERIC MATERIALS

The Effect of Molybdenum Disulfide on the Physical, Mechanical, and Technological Properties of Polyamide-6

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Abstract—The article considers the features of polyamide-6 modification with additives based on molybdenum disulfide, which significantly expands the scope of its application. However, polymer modification not only alters the target properties but may also affect other characteristics, sometimes negatively. The main task for developers of composite material is to select a balanced composition to achieve the desired goal—obtaining an antifriction, wear-resistant composite. The study presents a brief analysis of the effect of molybdenum disulfide on the overall properties of the polymer composites based on a literature review of publications from the last 5–7 years (16 sources, including cross-references). The effect of molybdenum disulfide on the physico-mechanical and technological properties of polyamide-6 has been investigated. In addition, various approaches to the creation of polymer composites are discussed. Initially, a fairly wide range of additive concentrations was studied and a narrow sector of its rational and effective use was determined. In the second stage of the study, the approach to polymer modification was changed. A modification scheme focused on large-scale production of polymer products was used. Based on the results of preliminary studies, international experience in this area was analyzed. It was noted that similar effects were observed when creating nanocomposites using modified molybdenum disulfide samples, but the general behavior of the polymer material and the achieved characteristics are different. The study highlights the challenges in producing MoS_2-based composites via direct mixing of components in a twin-screw extruder and proposes a method for their production using concentrates. The behavior of polymer compositions in the areas of ultimate deformations were investigated, revealing an increase in tensile strength and a change in the destruction mechanisim with hypotheses proposed regarding their nature. It was noted that the zone of ultimate deformations could be a subject of further research and is of interest in the context of forming highly oriented states in polyamide-6 composites.

Keywords: polymer materials, polyamide-6, polyamide modification, molybdenum disulfide

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INTRODUCTION

The wide variety of polymeric materials offers vast opportunities in designing products for various applications. However, each engineered product must meet specific performance requirements. In many cases, it is impossible to find a suitable commercially available polymer that meets the desired conditions. In such situations, modifying conventional materials with special additives and fillers provides an effective solution.

Today, general approaches and standard solutions have been developed for the targeted modification of polymer properties. This allows the improvement of specific material characteristics without significant degradation of other properties. For instance, additives such as graphite, molybdenum disulfide (MoS_2), ultra-high-molecular-weight polyethylene (UHMWPE), and silicone oils are widely used to alter the frictional properties of polymers. The use of modified polyamide-6 (PA-6) in metal-polymer friction pairs has also become widespread [1].

This study focuses on the application of molybdenum disulfide in PA-6 composites.

Table 1. Properties of polyamide-6 (food) produced in Grodno

Parameter	Value
View and color	Granules natural colors
Granule size (length/diameter), mm	2.2/2.3
Relative viscosity (sulfuric acid 96±0.15%, 25±0.1ºC), rel. units	3.31
Weight fraction of water (GOST 14870, cabinet method), %	0.04
Weight fraction of extractable substances (Soxhlet apparatus), %	0.5
Number of granules with foreign point inclusions per 100 g of product, pcs.	0
Number of oxidized granules in all samples selected for analysis (weight 2.0±0.01 kg), pcs.	0
Bulk density, g/cm³	0.63

Molybdenum disulfide (MoS_2), like graphite, has a layered structure. Each MoS_2 layer consists of stacked S–Mo–S sheets, just three atoms thick, with individual two-dimensional (2D) layers held together by van der Waals forces [2]. The weak interlayer interactions in MoS_2 facilitate easy sliding under applied load, resulting in minimal friction [3]. Due to this property, MoS_2 is used as a solid lubricant and an anti-friction filler in polymers [4], significantly enhancing the wear resistance of composites [5].

Studies show that PA-6/MoS_2 composites exhibit lower wear rates and improved product durability under dry friction conditions compared to unmodified PA-6 [6].

Despite the advantages of MoS_2, the addition of graphite to the PA-6 polymer matrix is often preferred due to its superior resistance to scuffing and lower coefficient of friction [7].

At the same time, MoS_2 acts as an effective lubricant in friction pairs even at elevated temperatures where conventional liquid lubricants fail [8].

The thermal stability of PA6/mMoS_2 composites also improves slightly. Tensile strength, Young's modulus, and elastic modulus increase compared to pure PA6 [9].

Some layered fillers enhance the material's ability to withstand high temperatures without losing mechanical integrity. This is particularly beneficial in high-friction and high-heat applications, such as automotive components, industrial machinery parts, and ultrasonic welding of polymer workpieces [10]. The increased thermal conductivity provided by MoS_2 also helps dissipate heat, reducing the risk of thermal degradation [11]. Notably, nanoscale MoS_2 in a PMMA matrix significantly improves the composit's thermal stability [12].

Enhanced mechanical wear resistance and thermal stability of PA6/MoS_2 composites make them suitable for various applications across multiple industries:

– Automotive components: Gears, bearings, and bushings where friction and wear reduction are critical;

– Industrial equipment: Wear-prone parts such as rollers and cams;

– Aerospace industry: Applications requiring materials to withstand high mechanical loads at elevated temperatures

Research Methodology

The positive effect of molybdenum disulfide on the frictional properties of polymer materials has been well documented in literature. However, the introduction of this additive also alters other characteristics of polymer composites, such as strength properties. This work was aimed at investigating the effect of molybdenum disulfide on the physico-mechanical properties of PA6. Understanding this effect will broaden the potential applications of this composite.

EXPERIMENTALS

Materials

The following materials were used in this study:

Base materials. Polyamide-6 (PA-6) TU RB 500048054.037-2002 (manufactured by JSC Grodno Azot, Belarus, technical specifications provided in Table 1; Polyamide-6 from Volgamid F34 food grade, TU 2224-040-00205311-08 (KuibyshevAzot, Russia).

Modifying additives. Molybdenum disulfide DMI-7, TU 48-19-133-90 (Russia, technical specifications provided in Table 2). Thermal stabilizer Irganox 1098 (CAS no. 23128-74-7. manufactured by BASF, Germany); Thermal stabilizer Songnox 1680 (CAS no. 31570-04-4. Songwon Industrial Co., Ltd., Korea).

Methods of Composite and Sample Preparation

To prepare polymer composites with varying contents of the modifying additive, different approaches were employed: Direct introduction of the additive during twin-screw compounding and introduction of the additive in the form of a concentrate based on the same polymer. Significant challenges in precise dosing and uniform distribution of the target additive—particularly at concentrations below 0.5%—necessitated the development of a 10% masterbatch of the target additive (molybdenum disulfide, MD) using the base material.

Masterbatch formulation (wt. parts): Polyamide-6 89.6, thermal stabilizer Irganox 1098 0.2, thermal stabilizer Songnox 1680 0.2, molybdenum disulfide DMI-7 10.

The production line for preparing the masterbatch using a Labtech LTE-20 laboratory twin-screw extruder is shown in Fig. 1.

Sample Preparation for Physico-Mechanical Testing

Samples were produced by injection molding on a Minijet MJ35 thermoplastic injection molding machine (CHEN HSONG, China) under the following processing conditions recommended by the polymer manufacturer (JSC Grodno Azot): Processing temperature 260–280°C, mold temperature 50°C. Prior to processing, the polymer was dried at 80°C for 2 h in accordance with the technical datasheet.

For each additive concentration, a series of 20 standard samples was produced. Transitional regime samples were excluded from the analysis. The samples for physico-mechanical testing complied with GOST (State standard) 11262-80, Type 1. The injection mold was manufactured in accordance with ISO 294-1 [13].

Research Methods

Physico-mechanical testing. Testing was performed using a HORIZON 5ST universal tensile testing machine (Tinius Olsen, UK) in accordance with GOST 11262-2017 (ISO 527-2:2012).

Melt pressure and temperature evaluation. Process parameters were recorded directly from the control systems of production machinery.

Injection pressure vs. additive content. The relationship between injection pressure and additive content was determined using a Minijet MJ35 injection molding machine (CHEN HSONG, China).

Table 2. Properties of MoS_2 of DMI-7 grade

Parameter	Value
Average particle size *D*(50), μm	2.3
Weight fraction of the base substance (MoS_2), %, not less than	98.0
Weight fraction of oxidized molybdenum, %, not more than	0.09
Weight fraction of iron (Fe), %, not more than	0.25
Weight fraction of SiO_2, %, not more than	0.02
Weight fraction of moisture, %, not more than	0.5
Weight fraction of insoluble substances, %, not more than	0.5
Dispersion up to 7 μm, %, not less than	98

Melt pressure variation was measured using a Labtech MP030 filter test system (Taiwan) equipped with a differential pressure sensor (no filter screens) to ensure unrestricted melt flow through the extruder.

Experimental Procedure

The study of molybdenum disulfide's effect on PA-6 properties was conducted in two stages:

In the stage 1 a broad concentration range from 0.5 to 5 wt % was examined. Due to difficulties in single-screw plastication, a twin-screw extruder was employed. Powdered PA-6 (F34 grade) was used as base material. This study stage provided a comprehensive understanding of the changes in the composit's physico-mechanical, operational, and processing properties.

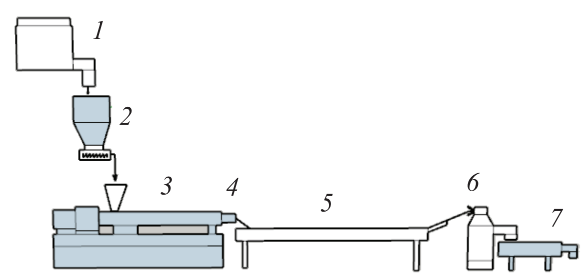

Fig. 1. Scheme of manufacturing DMC. (*1*) Mixer, (*2*) batcher, (*3*) twin screw extruder, (*4*) pelletizing head, (*5*) cooling bath, (*6*) cutting device, (*7*) classifier.

Table 3. Physic-mechanical properties of polyamide-6 and molybdenum disulfide composites

Sample no.	Composition	MoS_2 content, %	Tensile strength, MPa	Elongation at break, %
			test method (GOST 11262-80)	
1	PA-6	0	57.2 ± 4.3	153.0 ± 14.5
2	PA-6 +0.5% DMC	0.05	79.2 ± 5.26	201.0 ± 9.5
3	PA-6 +1.0% DMC	0.1	81.3 ± 4.2	206.0 ± 8.4
4	PA-6 +2.0% DMC	0.2	81.1 ± 1.9	206.0 ± 5.8
5	PA-6 +3.0% DMC	0.3	80.6 ± 5.0	208.0 ± 9.4
6	PA-6 +5.0% DMC	0.5	79.3 ± 0.3	203.0 ± 5.7

Stage 2 involved analyzing composite properties with target additive (molybdenum disulfide) content ranging from 0.05 to 0.5 wt %. Furthermore, this stage simulated industrial application of such additives, specifically: Modification of granulated polyamide-6 using masterbatches, implemented in standard processing technologies employing single-screw plastication (injection molding, extrusion, etc.) eliminating the use of costly twin-screw compounders.

This approach significantly simplifies production line equipment requirements and is directly applicable to high-volume manufacturing

RESULTS AND DISCUSSION

Investigation of the physico-mechanical properties of PA-6 modified with molybdenum disulfide

Our comprehensive study of polyamide-6 modified with molybdenum disulfide (MoS_2) across a wide concentration range revealed an anomalous improvement in mechanical properties: 20% increase in tensile strength, 24% enhancement in elongation at break. These improvements were observed across nearly all tested additive concentrations.

A comparative analysis with global research demonstrated that Chinese researchers [11] reported similar strength improvements in MoS_2-based nanocomposites. Notably, these effects of strength enhancement were detected at twice the concentration of modified MoS_2 compared to our study. Furthermore, the elongation at break values reported were more than an order of magnitude lower than our achieved results, accompanied by different sample fracture patterns. Similar behavior has been reported for polystyrene composites with MoS_2 adding below 0.05% [14].

In the second stage of the study we analyzed strength characteristics of composites incorporating the additive within precise narrow concentration range (0.05–0.5 wt %) using masterbatch incorporation method. The complete test results are presented in Table 3.

Notably, the incorporation of molybdenum disulfide into pure polyamide-6 at concentrations of 0.05, 0.1, 0.2, 0.3, and 0.5 wt % (in term of MoS_2 concentration) enhanced both tensile strength and elongation at break. Tensile strength increased by 20–40%. Elongation at break is improved by 24–35%. Mechanical properties do not show linear correlation with additive concentration. Maximum strength characteristics are achieved at just 0.1% MoS_2 content. Further increases in additive content provide no additional impact on mechanical properties but has a strong effect on the stability of technological processes, for example, during extrusion (reduction in the extrudate structural integrity, strand breakage, increased volatile emission from vent zones).

It is also worth noting the nature of sample fracturing during testing. Corresponding stress-strain curves are presented in Fig. 2.

It is obvious that all modified samples have a certain distinctive fracture behavior. Modified polyamide, when stretched to a 150% elongation, behaves identically to unmodified polymer. However, in the final plastic deformation stage, a dual-stage strengthening phenomenon is observed (Fig. 2). Such behavior is possible when the deformation-orientation mechanisms in the polymer structure change. This may be due to the effect of MoS_2 on the tribological properties of polyamide altering polymer chain orientation processes. Similar multi-stage strengthening is observed in repeatedly drawn polymer films and fibers under elevated temperatures [15].

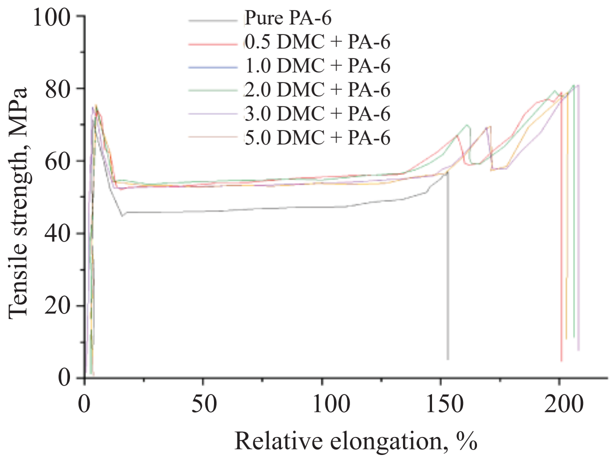

Fig. 2. Stress-strain curves of samples.

It is also notable that the final product properties are strongly influenced by processing-induced orientation (injection molding or extrusion) [16].

No significant melt pressure changes were observed during single-screw extrusion of MoS_2-containing compositions, which may be due to low shear rates and relatively small shear stress development. When producing samples by injection molding, the melt pressure slightly increases at MoS_2 concentrations from 0.05 to 0.2%.

When producing molybdenum disulfide masterbatch (10% MoS_2) and composites (0.5; 1.0; 2.0; 3.0, and 5.0% MoS_2) on a twin-screw extruder, strand pulsation was observed at low concentrations, indicating process instability. When the concentration increased to 5 and 10%, the process stabilized.

CONCLUSIONS

The study demonstrates the significant impact of molybdenum disulfide on the physico-mechanical properties of polyamide-6. An increase in mechanical strength by 20% or more is observed even at low MoS_2 concentrations. The peak of change in mechanical strength is observed at a 0.1% MoS_2 content. No significant gain in strength characteristics is observed with an increase in the additive concentration.

A change in the fracture behavior of the material at deformations close to the tensile strength limit is noted. Since the region of ultimate deformations is extremely interesting in light of the formation of highly oriented polymer structures, further research is required to understand the deformation mechanisms that occur in this composite.

The modified material exhibits unique failure mechanisms near ultimate tensile strength. Since the high-strain region is extremely interesting in the context of the formation of highly-oriented polymer structures, further research is required to understand the fracture mechanisms that occur in this composite.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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